

Building Classification From Aerial Imagery

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Abstract—This document is a template demonstrating the Ieee-conference format.

Index Terms—template; demo

I. INTRODUCTION

Travel forecasting models are based heavily on data provided by government statistical agencies, such as the US Census Bureau and the Bureau of Labor Statistics. These agencies provide the land use data needed to construct these models, giving statistics such as household number, median household income, etc. for census areas. In countries where statistical agencies are not as reliable, travel forecasting models are generally not developed or have data gaps. In countries where they are available, these models are highly reliant on the public release of such statistical data, which leads to problems if such survies are delayed or canceled.

Recent research has sought to find alternative methods to collect land use characterists. Gathering, analysing, and utilizing remote sensing data from satelites has been a major focus over the last several decades. Satellite aerial imagry has been used to analyze the volume and desnity man-made sturctures and roads to estiamte area populations, in both urban and rural settings [Allaw et al. [1]][2]. Day time imagery has also been used to track vehicle movement between major cities as an indication of economic activity [3]. It has also been used to classify land use by estimating landscape patterns and building fuctions, dividing different surfaces, and then categorizing building types [4]. Night time light data has also been shown to be useful as well. One study coorelated city light output with social economic variables such as population, GdP and electic power consumption [5], while another used the light emitted from cars at night on highways to predict vehicle travel between cities [6]. Satilite imagery has also be used to produce data sets such as the Microsoft Building Footprints data set, which has been used along with LiDAR and point of interest data to inventory residentila structures from census data [Microsoft [7]][8].

With the rise of commercially available and affordable aerial drones, the ability to capture remote sensing data such as high quality aerial imagery has become signifigantly easier. The

creation of open source softwares for drone use has also made it more accessible for use in research [9]. Though the pratical uses, economical fessibilty, and ethical concerns of collecting the higher resolution data is still being discussed [10].

More recently, with the advancement of more accessible deep learning models, machine learning has become very prelevant in analysing and classifying remote sensing data. One of it's primary uses so far has been to analyze large amounts of data, for such uses of tracking vehile movement between cities [3], or predicting population characterists and locations of man-made structures across entire countires. [2] Building classification is of particular interest for collecting land use characterists. Models such as "You Only Look Once" (YOLO) are being used to identify buildings and extract building footprints [Nurkarim and Wijayanto [11]][12].

Within this currently research the question arises "can remote sensing data be used to develop the land use data currently obtained from statistical agencies?" The devlopment of such a process or method would solve many problems in places without reliable statistidal data as well as suppliment data in other places.

In this paper the process of collecting building footprints for a city, annotating aerial images of the buildings, and training an YOLO model to classify building types is discusses. At the end it presents research and findings into creating a model that recongizes and categorizes building types based off remote sensing data, and discusses future research into estimating household statistics for a general area.

II. METHODOLGY

A. Data Collection

The city of Logan, Utah was chosen for analysis. Building footprints for Logan were obtained from the Microsoft Building Footprint public dataset. This dataset allowed for easy access to building footprint data for this inital study. All analysis was run using Python 3.12 and various python packages and APIs.

With the chosen city the latitude and longitude of the official city boundries was obtained. Using this boundry, all the building footprints in the city were extracted from the

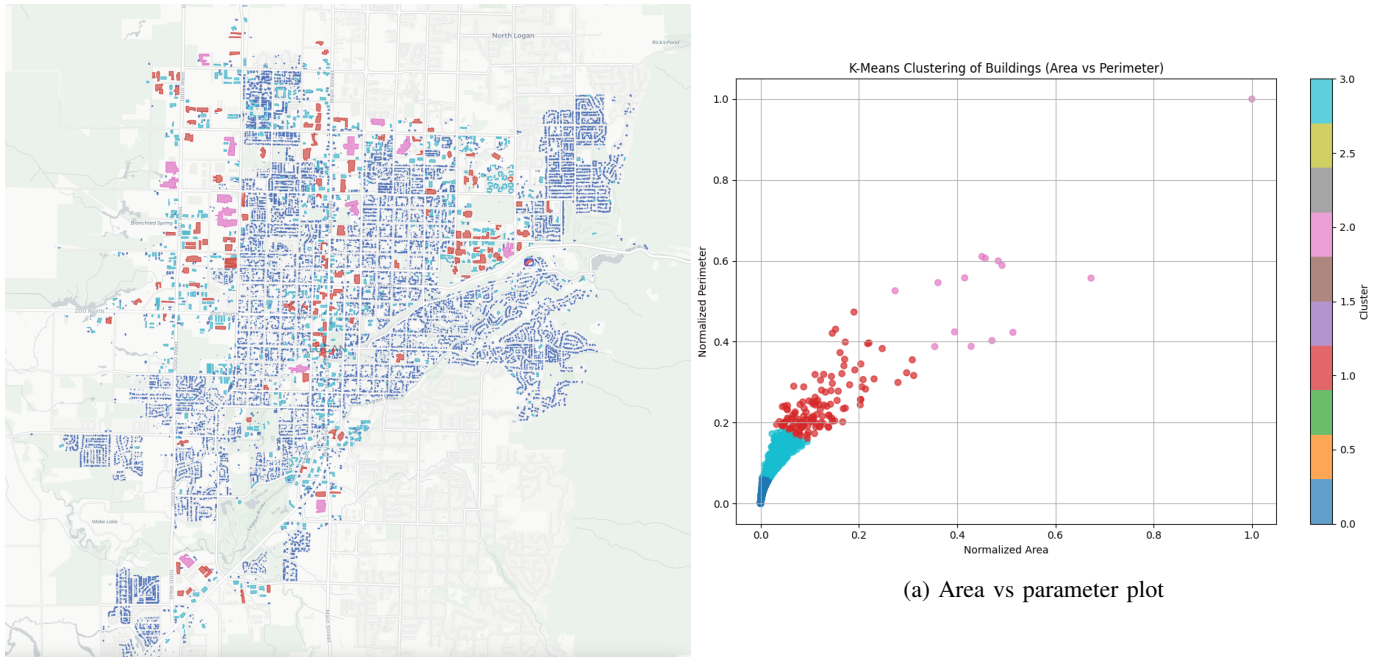


Figure 1: K-mean analysis for Logan, Utah

Microsoft Building Footprint dataset and saved as a geojson file. The building were then classified using a simple K-mean analysis to differentiate different sized buildings. To do this, the optimal K-mean was found using the elbow method. Using this K value, a K-mean analysis was run comparing the areas of the building footprints to their perimeters. Figure 1 shows the distribution of different cluster types, with most of the buildings in Logan being a smaller size.

Using the building footprint location and the python Library “Contextily”, 600 randomly chosen aerial images of building in Logan were taken from publicly available satellite imagery and cropped to focus on a single building. 200 of the images were from clusters that included only larger buildings to help balance building types in the dataset.

B. Model Training

Ultralytics’ “You Only Look Once” or YOLO model is a open source object detection model that has become very popular within do due it’s accuracy and ease of use. The model is trained on images that have been previously annotated, using bounding boxes, to identify the categories desired for the model to detect. The model trains itself on these images over many iterations until it is able to consistanly recongnize the different categories of objects. The YOLO model is then able to predict these categories in other images by bounding them in bounding boxes and displaying the probability of correct identification.

Using the open source tool Label Studio, the collected street map images were mannually annotated. This annotation including creating bounding boxes around the roof of each buliding, using best judgement based off the appearance of the building and surounding buildings and area. The main

buildings in each picture was labeld into five building categories: Apartment, Residential, Commercial, Church, and School. This order was consitant across annotations. These bounding boxes were saved as text files and each one matched up with its corresponding image.

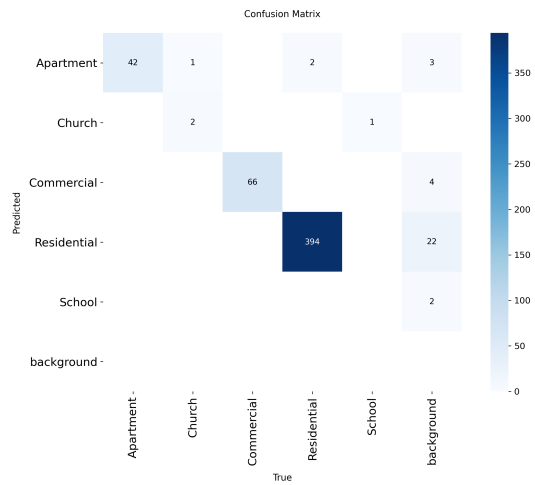
The annotated data sets were split into two categories. 80% of the images were used in training the object dection model, while 20% of the images were reserved for validation purposes. Within these image sets clusters that included larger and less common building sizes were specifically included, due to thier relatively lower likelihood of appearing, to make sure the model would be trained on these less common building types.

The model was run through hundreds of iterations until the model was able to accurety and reliably predict building types in Logan Utah. The results of the final validation of the model is shown in Figure 2.

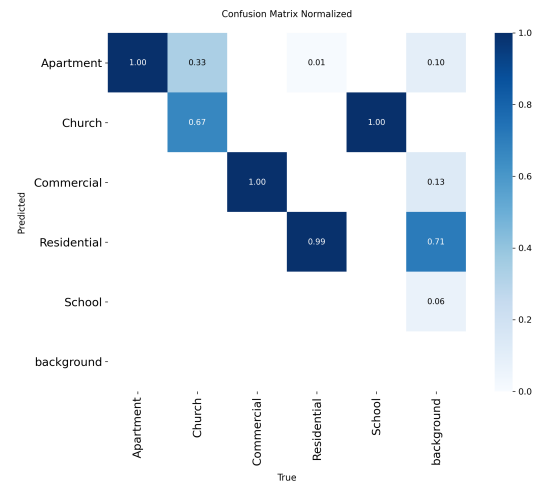
C. Model Validation and Comparison

Once the model was trained and validated using the Logan, Utah data set, the model was then tested upon other cities in Utah to assess it’s viability in predicting building categories elsewhere in Utah. Using American Community Survey data, five other test cities were chosen based off the city’s median income level, and the median age of the homes. The chosen cities an been seen in Table Ia.

Following the same method as previously described, building footprint data was collected and analized for each city. 200 aerial images of buildings were randomly chosen and annotated. These annotated data sets were then run through the model using validation only. The results of these runs are given below in Figure 3. These results were then analyses to



(a) Logan confusion matrix



(b) Logan confusion Matrix normalized

Figure 2: YOLO confusion matrixes for Logan, Utah.

Table I

(a) Chosen Cities Based off Median Income and House Age

House Age	High Income	Middle Income	Low Income
Old Houses	Bountiful, Utah	Ogden, Utah	Price, Utah
New Houses	Saratoga Springs	St. George, Utah	(None)

evaluate the ability of the model to predict in other cities, as well as assess if household income or house age were indicating factors.

III. RESULTS AND DISCUSSION

The Logan trained YOLO model was able to accurately predict residential, commercial, and apartment building types almost 100% of the time, as shown in Figure 2. The desired outcome is to have all the numbers in a diagonal line, which would indicate that the model is predicting the category of a building that is in line with the annotations made.

The model did underperform with schools and churches. This is attributed to the fact that there were very low number of images of both building types provided during model training.

A important aspect to discuss is the model predicting different building types where there is no building. This is seen on the far right of each graph, where the background was categorized. This error occurs from two sources. One is the annotation. Building such as backyard sheds were not identified, but the model still picked them up. Or there were times that only a fourth of a building was present in an image, and so was not annotated, but was still recognized by the model at a lower probability level. The other source of error is the quality of the photos used. The satellite images from Contextily were convenient and easy to access, but were not high quality. This resulted in the model sometimes interpreting things of solid colors, such as roads and covered parking as

a type of building. The second error could be solved using higher quality photos, while the first

Confusion matrix for other analysis cities. What are the results? Was there a difference between how the model did between the cities? Was age of the house a big factor? Was income? Is there something else? Neither house age or house median income made a difference in the classification of houses.

Given the results of the validations, a process to develop a model for different locations would be more useful and viable than to create a universal model used in a wide area.

However, much of this research is susceptible to similar problems as transportation models. Satellite remote sensing data is not consistent around the world, having varying availability and image quality.

IV. CONCLUSION AND FUTURE RESEARCH

The conclusion goes here.

As discussed by Hall et al., there are potential ethical problems with such data collection. It straddles the line between public and private information. This will need to be considered as further research is done into this area as to stay on the right side of the law.[10]

What does the future research look like?

Training models to detect other things such as cars, swing sets, patios, etc.

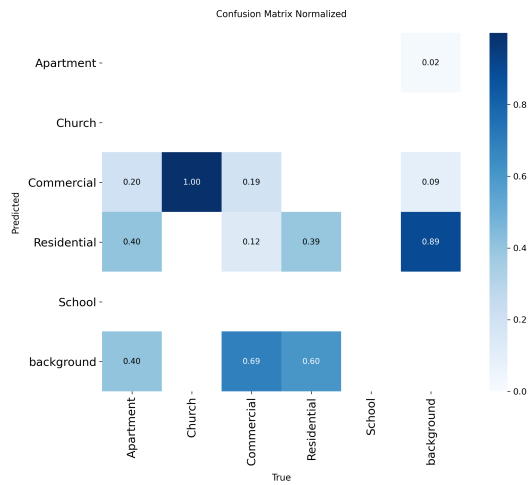
Creating the process to predict household types based off the collected data

Incorporating drone aerial imagery and extracting building footprints from that.

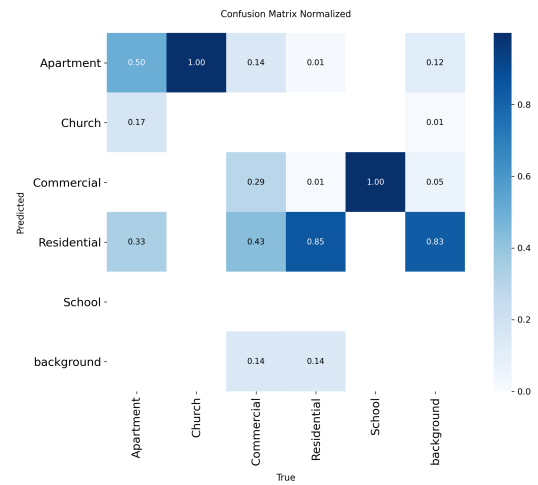
Research into other remote sensing data and how it could be incorporated.

V. ACKNOWLEDGMENT

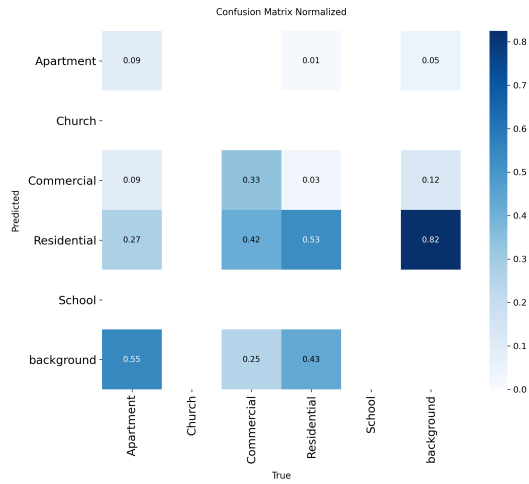
The authors would like to thank...



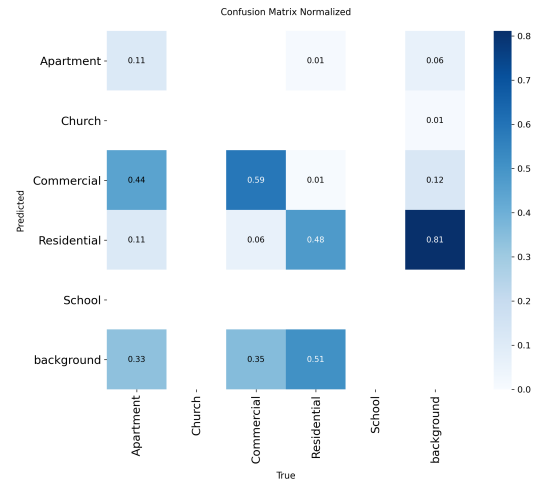
(a) Bountiful, Utah



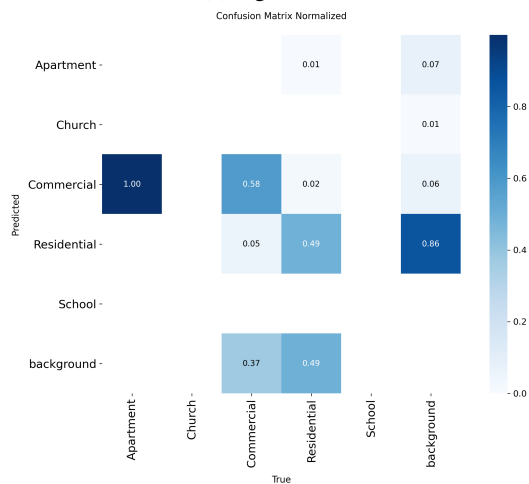
(b) Saratoga Springs, Utah



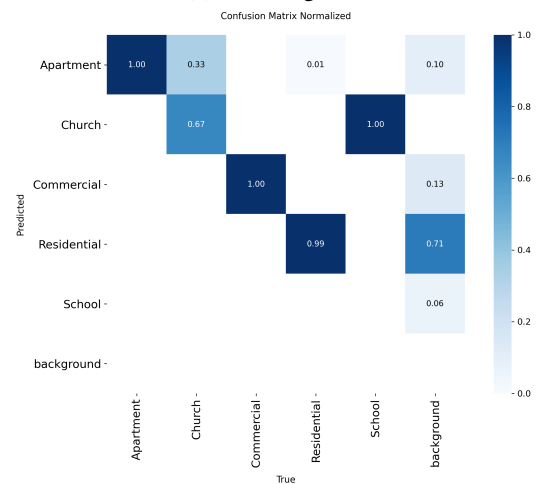
(c) Ogden, Utah



(d) St. George, Utah



(e) Price, Utah



(f) Logan Utah

Figure 3: Normalized YOLO confusion matrixes

REFERENCES

- [1] Kamel Allaw et al. "Population estimation using geographic information system and remote sensing for unorganized areas". In: *Geoplanning: Journal of Geomatics and Planning* 7.2 (Nov. 1, 2020), pp. 75–86.
- [2] Casper Samsø Fibæk, Carsten Keßler, and Jamal Jokar Arsanjani. "A multi-sensor approach for characterising human-made structures by estimating area, volume and population based on sentinel data and deep learning". In: *International Journal of Applied Earth Observation and Geoinformation* 105 (Dec. 25, 2021), p. 102628.
- [3] Patrick Lehnert et al. "Proxying economic activity with daytime satellite imagery: filling data gaps across time and space". In: *PNAS Nexus* 2.4 (Apr. 1, 2023), pgad099.
- [4] Ye Zhang et al. "Landscape patterns and building functions for urban land-use classification from remote sensing images at the block level: A case study of wuchang district, wuhan, china". In: *Remote Sensing* 12.11 (Jan. 2020), p. 1831.
- [5] Shankar Acharya Kamarajugedda, Pradeep V Mandapaka, and Edmond Y.M Lo. "Assessing urban growth dynamics of major southeast asian cities using nighttime light data". In: *International Journal of Remote Sensing* 38.21 (Nov. 2, 2017), pp. 6073–6093.
- [6] Ying Chang et al. "A novel method of evaluating highway traffic prosperity based on nighttime light remote sensing". In: *Remote Sensing* 12.1 (Jan. 2020), p. 102.
- [7] Microsoft. *GlobalMLBuildingFootprints*. Version Python. Jan. 7, 2026. URL: <https://github.com/microsoft/GlobalMLBuildingFootprints> (visited on 01/07/2026).
- [8] Xinyue Ye et al. "Enhancing population data granularity: A comprehensive approach using lidar, POI, and quadratic programming". In: *Cities* 152 (Sept. 2024), p. 105223.
- [9] Gustavo Pereyra Irujo et al. "Open science drone toolkit: open source hardware and software for aerial data capture". In: *PLOS ONE* 18.4 (Apr. 6, 2023), e0284184.
- [10] Ola Hall and Ibrahim Wahab. "The use of drones in the spatial social sciences". In: *Drones* 5.4 (Dec. 2021), p. 112.
- [11] Wahidya Nurkarim and Arie Wahyu Wijayanto. "Building footprint extraction and counting on very high-resolution satellite imagery using object detection deep learning framework". In: *Earth Science Informatics* 16.1 (Mar. 1, 2023), pp. 515–532.
- [12] Mayank Sharma and Rahul Dev Garg. "Building footprint extraction from aerial photogrammetric point cloud data using its geometric features". In: *Journal of Building Engineering* 76 (Oct. 1, 2023), p. 107387.